

Chapter 09 Part 2: Rate Law

1.



A sample of N_2O_5 was placed in an evacuated container, and the reaction represented above occurred. The value of $P_{\text{N}_2\text{O}_5}$, the partial pressure of $\text{N}_2\text{O}_5(\text{g})$, was measured during the reaction and recorded in the table below.

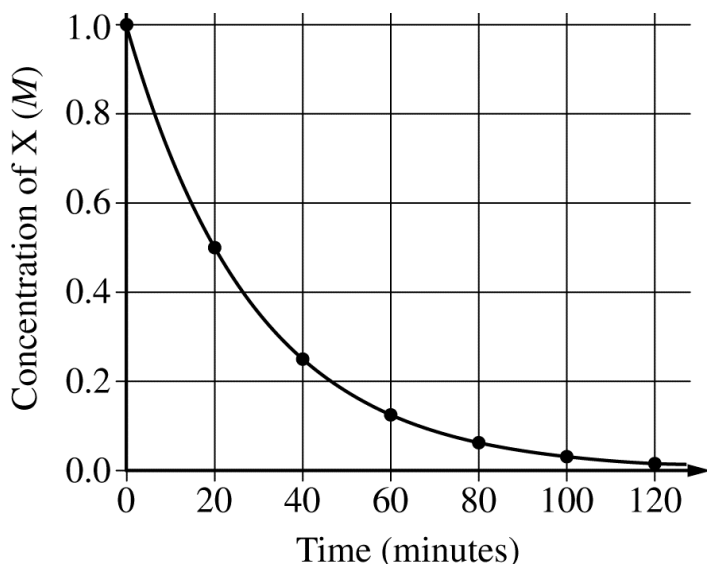
Time (min)	$P_{\text{N}_2\text{O}_5}$	$\ln(P_{\text{N}_2\text{O}_5})$	$\frac{1}{P_{\text{N}_2\text{O}_5}}$ (atm^{-1})
0	150	5.0	0.0067
100	75	4.3	0.013
200	38	3.6	0.027
300	19	2.9	0.053

Which of the following correctly describes the reaction?

- (A) The decomposition of N_2O_5 is a zero-order reaction.
- (B) The decomposition of N_2O_5 is a first-order reaction. ✓
- (C) The decomposition of N_2O_5 is a second-order reaction.
- (D) The overall reaction order is 3.

2. $\text{X} \rightarrow \text{Products}$

A student studied the kinetics of the reaction represented above by measuring the concentration of the reactant, X, over time. The data are plotted in the graph below.



Which of the following procedures will allow the student to determine the rate constant, k , for the reaction?

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- (A) Plot $\ln [X]$ versus time and determine the magnitude of the slope. ✓
- (B) Plot $1/[X]$ versus time and determine the magnitude of the slope.
- (C) Run another trial of the experiment with a different initial concentration, plot the data on the same graph, and see where the curves intersect.
- (D) Run another trial of the experiment at a different temperature, plot the data on the same graph, and see where the curves have the same slope.

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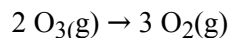
3. Answer the following questions about ozone

- a. The O_3 molecule has a central oxygen atom bonded to two outer oxygen atoms that are not bonded to one another. In the box below, draw the Lewis electron-dot diagram of the O_3 molecule. Include all valid resonance structures.



- b. Based on the diagram you drew in part (a), what is the shape of the ozone molecule?

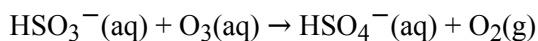
Ozone decomposes according to the reaction represented below



$$\Delta H^\circ = -285 \text{ kJ/mol}_{\text{rxn}}$$

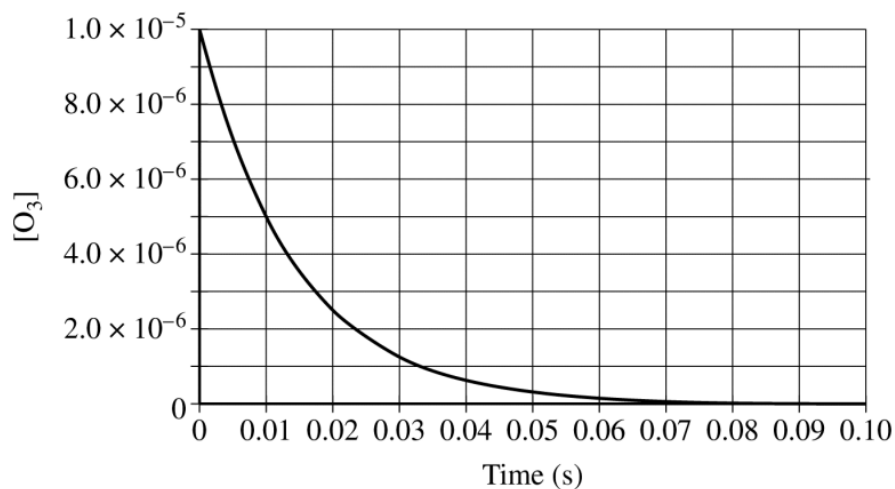
- c. The bond enthalpy of the oxygen-oxygen bond in O_2 is 498 kJ/mol. Based on the enthalpy of the reaction represented above, what is the average bond enthalpy, in kJ/mol, of an oxygen-oxygen bond in O_3 ?

Ozone can oxidize $\text{HSO}_3^-(\text{aq})$, as represented by the equation below.



A solution is prepared in which the initial concentration of $\text{HSO}_3^-(\text{aq})$ ($6.4 \times 10^{-4} \text{ M}$) is much larger than that of $\text{O}_3(\text{aq})$ ($1.0 \times 10^{-5} \text{ M}$). The concentration of $\text{O}_3(\text{aq})$ is monitored as the reaction proceeds, and the data are plotted in the graph below

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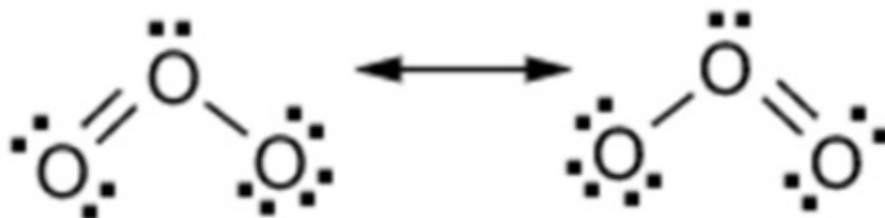
- d. The data are consistent with the following rate law: $\text{rate} = k_1[\text{O}_3]$
- Based on the graph on the previous page, determine the half-life of the reaction
 - Determine the value of the rate constant, k_1 , for the rate law. Include units with your answer.
 - Considering the relative concentrations of the reactants, briefly explain why the data in the graph are also consistent with the following rate law: $\text{rate} = k_2[\text{O}_3][\text{HSO}_3^-]$
 - Briefly describe an experiment that could provide evidence to support the rate law given in part (d)(iii)

Part A

point(s) maximum

1 point is earned for a diagram that includes all 18 valence electrons and obeys the octet rule.

1 point is earned for showing correct application of resonance.



✓

0

1

2

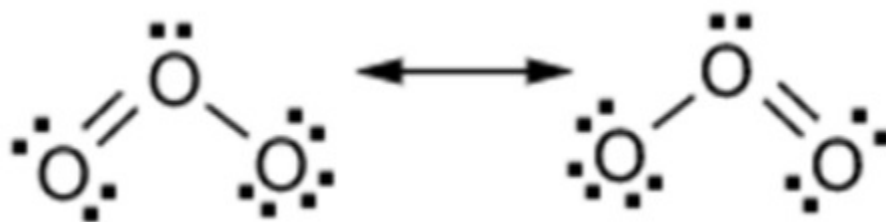
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Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for a diagram that includes all 18 valence electrons and obeys the octet rule.

1 point is earned for showing correct application of resonance.

**Part B****1 point(s) maximum**

1 point is earned for the shape based on student's Lewis diagram.

The ozone molecule has a bent shape.



0

1

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the shape based on student's Lewis diagram.

The ozone molecule has a bent shape.

Part C**2 point(s) maximum**

1 point is earned for the correct determination of the number of bonds broken in the reactants and the number of bonds formed in the products.

$$\Delta H^\circ = \Sigma (\text{bond enthalpies})_{\text{reactants}} - \Sigma (\text{bond enthalpies})_{\text{products}}$$

Four equivalent bonds are broken in two O₃ molecules. Three oxygen-oxygen bonds are formed in three O₂ molecules.

1 point is earned for the calculation of the energy of the bonds in O₃ consistent with bond counting, the bond energy of

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O₂, and the $\Delta H^\circ_{\text{rxn}}$.

Let x = the bond enthalpy in ozone, therefore

$$4x - 3(498 \text{ kJ/mol}) = -285 \text{ kJ/molrxn}$$

$$x = 302 \text{ kJ/mol}$$



0	1	2
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Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for the correct determination of the number of bonds broken in the reactants and the number of bonds formed in the products.

$$\Delta H^\circ = \Sigma (\text{bond enthalpies})_{\text{reactants}} - \Sigma (\text{bond enthalpies})_{\text{products}}$$

Four equivalent bonds are broken in two O₃ molecules. Three oxygen-oxygen bonds are formed in three O₂ molecules.

1 point is earned for the calculation of the energy of the bonds in O₃ consistent with bond counting, the bond energy of O₂, and the $\Delta H^\circ_{\text{rxn}}$.

Let x = the bond enthalpy in ozone, therefore

$$4x - 3(498 \text{ kJ/mol}) = -285 \text{ kJ/molrxn}$$

$$x = 302 \text{ kJ/mol}$$

Part D

5 point(s) maximum

Part (i)

1 point is earned for the correct answer.

$$\text{Half-life} = 0.010 \text{ s}$$

Part (ii)

1 point is earned for the correct value.

1 point is earned for the correct unit.

$$t_{1/2} = \frac{0.693}{k_1} \Rightarrow k_1 = \frac{0.693}{t_{1/2}} = \frac{0.693}{0.010 \text{ s}} = 69 \text{ s}^{-1}$$

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Part (iii)

1 point is earned for a correct explanation.

The data are consistent with rate = $k_2 [\text{O}_3][\text{HSO}_3^-]$ because $[\text{HSO}_3^-]$ remains essentially constant during the experiment. ($[\text{HSO}_3^-]$ can be contained in the rate constant k_1 .)

Part (iv)

1 point is earned for describing a valid experiment.

Repeat the experiment with a different initial concentration of HSO_3^- . (If the change in the rate of the reaction is directly proportional to the change in $[\text{HSO}_3^-]$, then the reaction is first order with respect to HSO_3^- .)



0	1	2	3	4	5
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Student response earns 5 of the following 5 point(s)

5 point(s) maximum

Part (i)

1 point is earned for the correct answer.

Half-life = 0.010 s.

Part (ii)

1 point is earned for the correct value.

1 point is earned for the correct unit.

$$t_{1/2} = \frac{0.693}{k_1} \Rightarrow k_1 = \frac{0.693}{t_{1/2}} = \frac{0.693}{0.010 \text{ s}} = 69 \text{ s}^{-1}$$

Part (iii)

1 point is earned for a correct explanation.

The data are consistent with rate = $k_2 [\text{O}_3][\text{HSO}_3^-]$ because $[\text{HSO}_3^-]$ remains essentially constant during the experiment. ($[\text{HSO}_3^-]$ can be contained in the rate constant k_1 .)

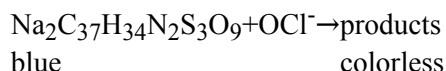
Part (iv)

1 point is earned for describing a valid experiment.

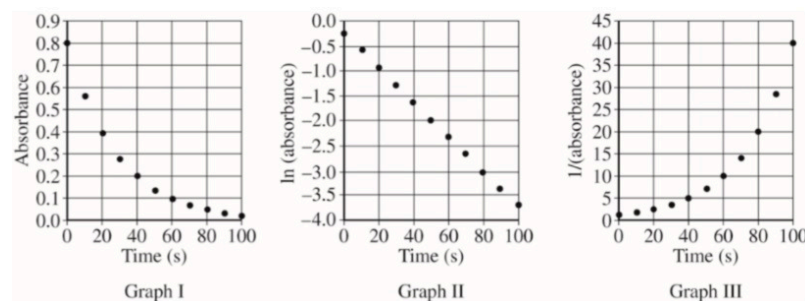
Repeat the experiment with a different initial concentration of HSO_3^- . (If the change in the rate of the reaction is directly proportional to the change in $[\text{HSO}_3^-]$, then the reaction is first order with respect to HSO_3^- .)

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4.



Blue food coloring can be oxidized by household bleach (which contains OCl^-) to form colorless products, as represented by the equation above. A student used a spectrophotometer set at a wavelength of 635 nm to study the absorbance of the food coloring over time during the bleaching process. In the study, bleach is present in large excess so that the concentration of OCl^- is essentially constant throughout the reaction. The student used data from the study to generate the graphs below.



- a. Based on the graphs above, what is the order of the reaction with respect to the blue food coloring?
- b. The reaction is known to be first order with respect to bleach. In a second experiment, the student prepares solutions of food coloring and bleach with concentrations that differ from those used in the first experiment. When the solutions are combined, the student observes that the reaction mixture reaches an absorbance near zero too rapidly. In order to correct the problem, the student proposes the following three possible modifications to the experiment.
- Increasing the temperature
 - Increasing the concentration of the food coloring
 - Increasing the concentration of the bleach
- Circle the one proposed modification above that could correct the problem, and explain how that modification increases the time for the reaction mixture to reach an absorbance near zero.
- c. In another experiment, a student wishes to study the oxidation of red food coloring with bleach. How would the student need to modify the original experimental procedure to determine the order of the reaction with respect to the red food coloring?

Part A

1 point is earned for the correct order.

First order

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0	1
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The student response earns one of the following points:

1 point is earned for the correct order.

First order

Part B

1 point is earned for the correct choice.

["Increasing the concentration of the food coloring" should be circled.]

1 point is earned for a correct explanation.

If the initial concentration of blue food coloring is increased, then more time is required (regardless of the reaction order indicated in part (a)) for the bleach to oxidize the additional blue food coloring.



0	1	2
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The student response earns two of the following points:

1 point is earned for the correct choice.

["Increasing the concentration of the food coloring" should be circled.]

1 point is earned for a correct explanation.

If the initial concentration of blue food coloring is increased, then more time is required (regardless of the reaction order indicated in part (a)) for the bleach to oxidize the additional blue food coloring.

Part C

1 point is earned for a correct answer.

The spectrophotometer should be set to a different wavelength.



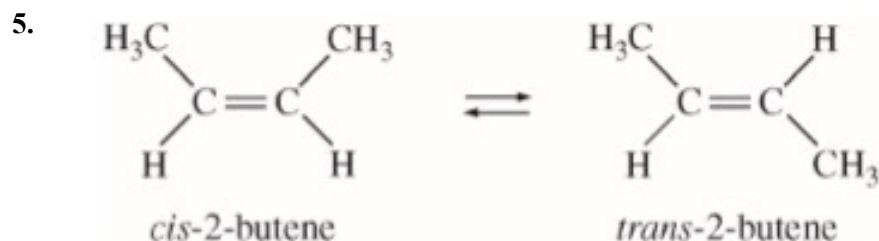
0	1
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The student response earns one of the following points:

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1 point is earned for a correct answer.

The spectrophotometer should be set to a different wavelength.



The half-life ($t_{1/2}$) of the catalyzed isomerization of *cis*-2-butene gas to produce *trans*-2-butene gas, represented above, was measured under various conditions, as shown in the table below.

Trial Number	Initial $P_{\text{cis-2-butene}}$ (torr)	V (L)	T (K)	$t_{1/2}$ (s)
1	300.	2.00	350.	100.
2	600.	2.00	350.	100.
3	300.	4.00	350.	100.
4	300.	2.00	365	50.

- The reaction is first order. Explain how the data in the table are consistent with a first-order reaction.
- Calculate the rate constant, k , for the reaction at 350. K. Include appropriate units with your answer.
- Is the initial rate of the reaction in trial 1 greater than, less than, or equal to the initial rate in trial 2 ? Justify your answer.
- The half-life of the reaction in trial 4 is less than the half-life in trial 1. Explain why, in terms of activation energy.

Part A

1 point is earned for a correct explanation.

For a first-order reaction, the half-life is independent of reactant concentration (or pressure) at constant T , as shown in trials 1, 2, and 3.



0	1
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The student response earns one of the following points:

1 point is earned for a correct explanation.

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For a first-order reaction, the half-life is independent of reactant concentration (or pressure) at constant T , as shown in trials 1, 2, and 3.

Part B

1 point is earned for correct numerical answer with units.

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{100. \text{ s}} = 0.00693 \text{ s}^{-1}$$



0	1
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The student response earns one of the following points:

1 point is earned for correct numerical answer with units.

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{100. \text{ s}} = 0.00693 \text{ s}^{-1}$$

Part C

1 point is earned for the correct answer with justification.

The initial rate in trial 1 is less than that in trial 2 because rate = k [*cis*-2-butene] or rate = k *Pcis*-2-butene (with reference to values from both trials).

OR

because the initial concentration of *cis*-2-butene in trial 1 is less than that in trial 2 and k is constant.



0	1
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The student response earns one of the following points:

1 point is earned for the correct answer with justification.

The initial rate in trial 1 is less than that in trial 2 because rate = k [*cis*-2-butene] or rate = k *Pcis*-2-butene (with reference to values from both trials).

OR

because the initial concentration of *cis*-2-butene in trial 1 is less than that in trial 2 and k is constant.

Part D

1 point is earned for a correct answer with justification.

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The temperature is higher in trial 4, meaning that the KE_{avg} of the molecules is greater. Consequently, in this trial a greater fraction of collisions have sufficient energy to overcome the activation energy barrier, thus the rate is greater.



0	1
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The student response earns one of the following points:

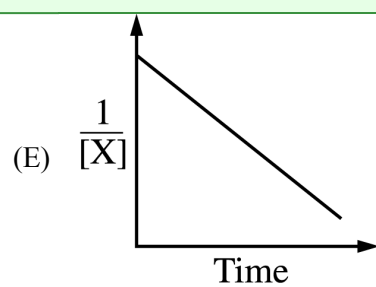
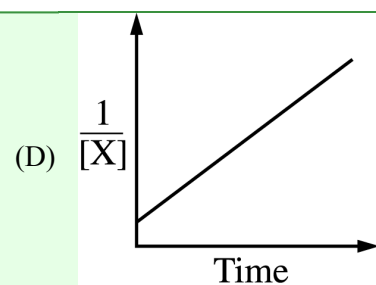
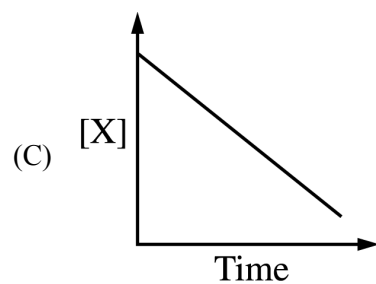
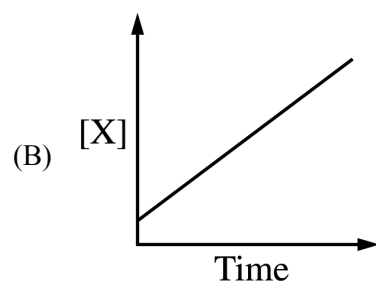
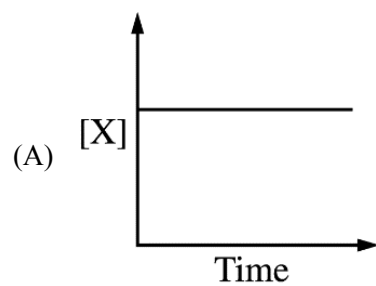
1 point is earned for a correct answer with justification.

The temperature is higher in trial 4, meaning that the KE_{avg} of the molecules is greater. Consequently, in this trial a greater fraction of collisions have sufficient energy to overcome the activation energy barrier, thus the rate is greater.

6. $X \rightarrow \text{products}$

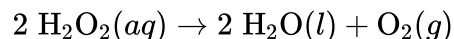
Pure substance X decomposes according to the equation above. Which of the following graphs indicates that the rate of decomposition is second order in X ?

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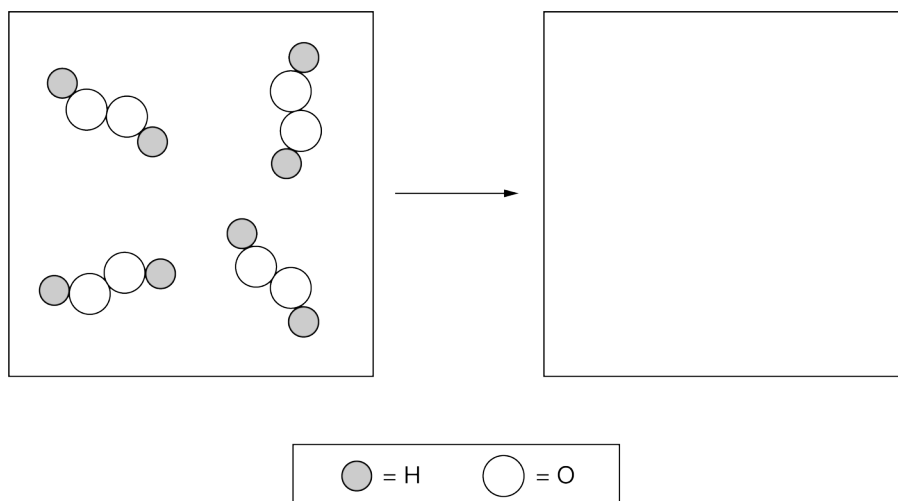
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7. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

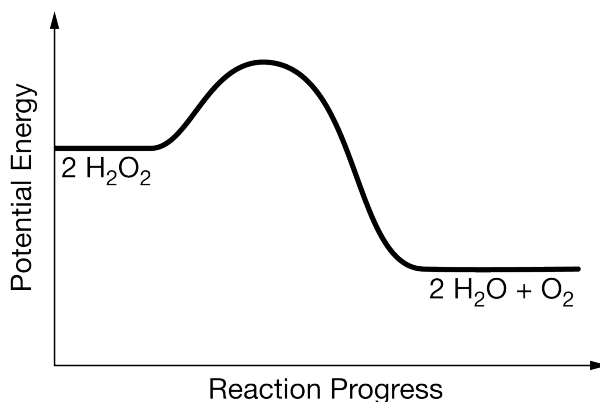


Hydrogen peroxide, H_2O_2 , decomposes according to the equation above. This reaction is thermodynamically favorable at room temperature.

- (a) A particulate representation of the reactants is shown below in the box on the left. In the box below on the right, draw the particulate representation of all the molecules that would be produced from these four reactant molecules.



- (b) Shown below is a potential energy diagram for the uncatalyzed decomposition of $\text{H}_2\text{O}_2(aq)$.

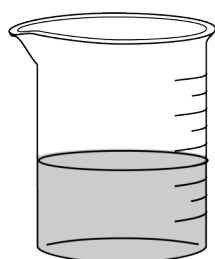


- (i) According to the diagram, is the decomposition reaction exothermic or endothermic? Justify your answer.
- (ii) Manganese dioxide, $\text{MnO}_2(s)$, is an insoluble substance that acts as a catalyst for the decomposition

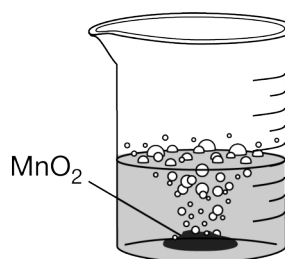
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reaction. On the diagram above, draw a curve to represent the reaction as it occurs in the presence of $\text{MnO}_2(s)$.

A student investigates the decomposition reaction in the laboratory. The student prepares two small beakers, adding 20.0 mL of 9.77 M $\text{H}_2\text{O}_2(aq)$ to each one. Each beaker is placed on an electronic balance. The student adds 0.10 g of $\text{MnO}_2(s)$ to the second beaker and records the mass of each beaker and its contents at 10-second intervals for one minute. The beakers and the data are shown below.



Beaker 1



Beaker 2

Time (seconds)	Mass of Beaker 1 + $\text{H}_2\text{O}_2(aq)$ (grams)	Mass of Beaker 2 + $\text{H}_2\text{O}_2(aq)$ + $\text{MnO}_2(s)$ (grams)
0	43.09	43.19
10	43.09	43.06
20	43.09	42.94
30	43.09	42.83
40	43.09	42.73
50	43.09	42.65
60	43.09	42.58

(c) For beaker 2 during the 60-second period, calculate the following.

(i) The number of moles of $\text{O}_2(g)$ that was produced

(ii) The mass of $\text{H}_2\text{O}_2(aq)$ that decomposed

(d) The student continues the experiment for an additional minute. For beaker 2, will the mass of $\text{H}_2\text{O}_2(aq)$ consumed during the second minute be greater than, less than, or equal to the mass of $\text{H}_2\text{O}_2(aq)$ consumed during the first minute? Explain your answer referring to the data in the table.

(e) Based on the data, the student claims that the catalyzed reaction has zeroth-order kinetics. Do you agree with the student's claim? Justify your answer.

(f) A second student did the experiment using larger volumes of 9.77 M $\text{H}_2\text{O}_2(aq)$. The student noticed that the reaction in beaker 2 proceeded extremely rapidly, causing some of the liquid to splash out of the beaker onto the lab table. The student claims that as a result of the loss of the liquid from the beaker, the calculated number of

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moles of $\text{O}_2(g)$ produced is greater than the actual number of moles of $\text{O}_2(g)$ produced during the first 60 seconds. Do you agree with the student? Justify your answer.

(g) The hydrogen peroxide used in this experiment can be prepared by the reaction of solid ammonium peroxydisulfate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$, with water. The products are hydrogen peroxide (H_2O_2) and ammonium bisulfate (NH_4HSO_4). Write the balanced equation for the reaction.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

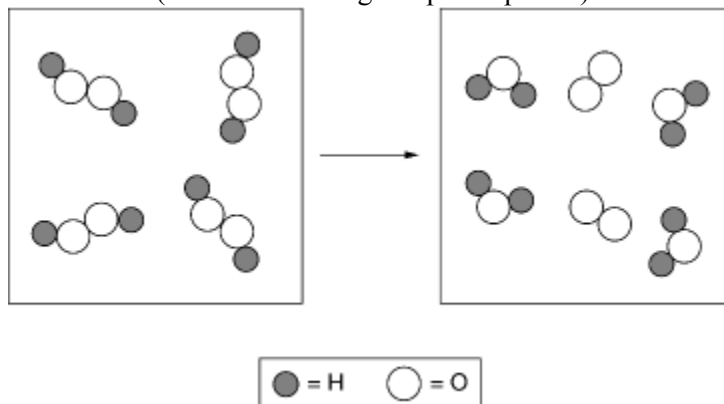


0	1	2
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The response meets both of the following criteria.

- ☐ The diagram shows H_2O molecules and O_2 molecules.
- ☐ The diagram shows atom balance.

(See the following sample response.)



Part B(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ The response indicates that the reaction is exothermic.

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- ☐ The response gives a justification such as the following.
The products have lower potential energy than the reactants. Thus, the system has lost potential energy, releasing heat.

Part B(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

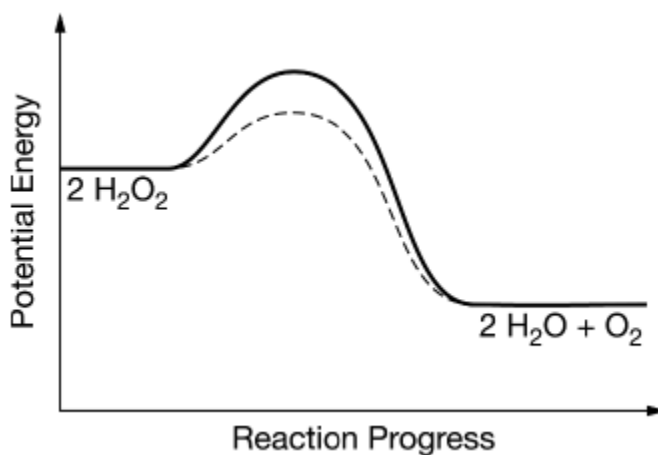


0	1
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The response shows a curve for the catalyzed reaction that meets both of the following criteria.

- ☐ The reactants and products are at the same energy levels as the original curve
- ☐ The activation energy is lower than that of the original curve.

(See the following sample response.)

**Part C(i)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives calculations equivalent to the following.

$$\text{mass O}_2 = 43.19 \text{ g} - 42.58 \text{ g} = 0.61 \text{ g}$$

$$\text{moles O}_2 = 0.61 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{32.00 \text{ g}} = 0.019 \text{ mol O}_2$$

Chapter 09 Part 2: Rate Law**Part C(ii)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If an incorrect answer to part (c)(i) is used correctly in part (c)(ii), full credit can be earned in part (c)(ii).



0	1
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The response gives a calculation equivalent to the following.

$$\text{mass H}_2\text{O}_2 = 0.019 \text{ mol O}_2 \times \frac{2 \text{ mol H}_2\text{O}_2}{1 \text{ mol O}_2} \times \frac{34.02 \text{ g H}_2\text{O}_2}{1 \text{ mol H}_2\text{O}_2} = 1.3 \text{ g H}_2\text{O}_2$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ The response indicates that the mass of $\text{H}_2\text{O}_2(aq)$ consumed during the second minute would be less than the mass of $\text{H}_2\text{O}_2(aq)$ consumed during the first minute.
- ☐ The response indicates that the decrease in mass becomes smaller over every subsequent 10-second interval and this trend would continue in the second minute.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ The response indicates “disagree”.
- ☐ The response indicates that if the reaction were zeroth-order, the rate of mass loss (which corresponds to the rate of reaction) would be constant until the reactant was consumed, but the data show that the rate of mass loss from beaker 2 is not constant.

Part F

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ The response indicates agreement.
- ☐ The response indicates that the calculated amount of O_2 produced is based on the change in mass, and in this case some of the change in mass is due to loss of solution rather than production of O_2 .

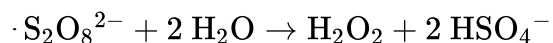
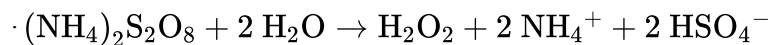
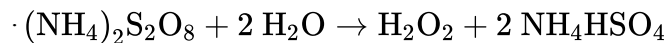
Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

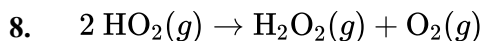


0	1
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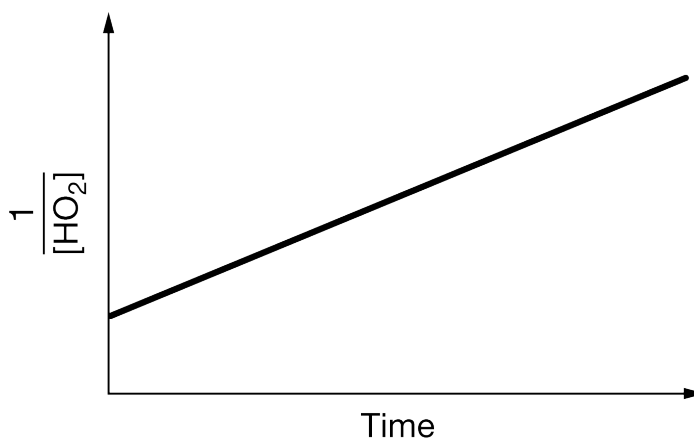
The response gives one of the following equations.



Chapter 09 Part 2: Rate Law



The reaction represented by the chemical equation shown above occurs in Earth's atmosphere. In an experiment, $[\text{HO}_2]$ was monitored over time and the data plotted as shown in the following graph.



Based on the information, which of the following is the rate law expression for the reaction?

(A) $\text{Rate} = k$

(B) $\text{Rate} = k[\text{HO}_2]$

(C) $\text{Rate} = k[\text{HO}_2]^2$

(D) $\text{Rate} = k[\text{H}_2\text{O}_2][\text{O}_2]$



9. After a certain pesticide compound is applied to crops, its decomposition is a first-order reaction with a half-life of 56 days. What is the rate constant, k , for the decomposition reaction?

(A) 0.012 day^{-1}

(B) 0.018 day^{-1}

(C) 56 day^{-1}

(D) 81 day^{-1}



10. After a certain pesticide compound is applied to crops, its decomposition is a first-order reaction with a half-life of 32.5 days. What is the rate constant, k , for the decomposition reaction?

(A) 0.0213 day^{-1}

(B) 0.0308 day^{-1}

(C) 32.5 day^{-1}

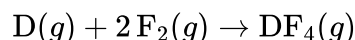
(D) 46.9 day^{-1}



Chapter 09 Part 2: Rate Law

11. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

An unknown gas, D, reacts with fluorine gas to form the compound $\text{DF}_4(g)$, as represented by the following equation.



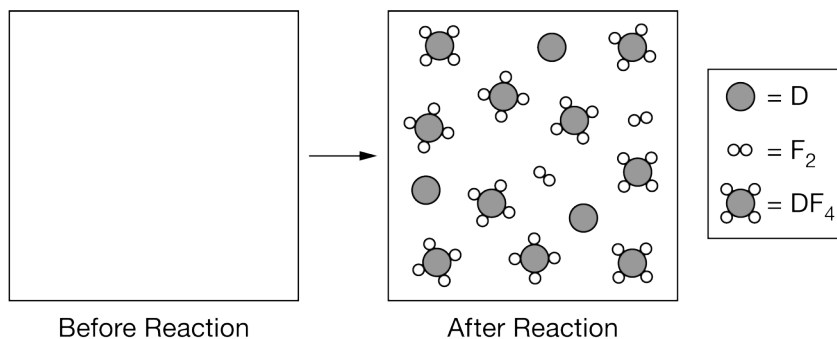
A series of experimental trials were performed at 600°C to determine the rate law expression for the reaction. Data from the trials are shown below.

	Initial P_{D} (torr)	Initial P_{F_2} (torr)	Initial Rate of Appearance of $\text{DF}_4(g)$ (torr/min)
Trial 1	1500	1500	5.00×10^{-3}
Trial 2	3000	1500	10.0×10^{-3}
Trial 3	1500	3000	20.0×10^{-3}

- (a) A student claims that the reaction is first order with respect to $\text{F}_2(g)$. Do you agree or disagree? Justify your answer using the data above.
- (b) What is the initial rate of disappearance of $\text{F}_2(g)$ in trial 3? Justify your answer.
- (c) A student claims that if element D is in group 18, then the molecular geometry of $\text{DF}_4(g)$ is square planar. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\text{D}(g)$ and $\text{F}_2(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to “Before Reaction” is intentionally left blank.)

Chapter 09 Part 2: Rate Law

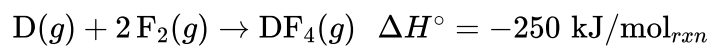


- (d) If the total pressure at the end of the reaction is 12 atm, what is the partial pressure of $D(g)$?
- (e) Assuming that only D and F_2 are present in the container before the reaction, determine the number of moles of D and F_2 that were present in the initial reaction mixture.

$D =$ _____

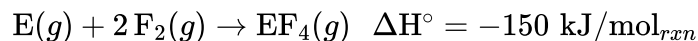
$F_2 =$ _____

The equation and enthalpy change for the reaction between $D(g)$ and $F_2(g)$ are given below.



- (f) If 0.50 mol of $DF_4(g)$ is produced when equimolar amounts of $D(g)$ and $F_2(g)$ react, how much heat is released?

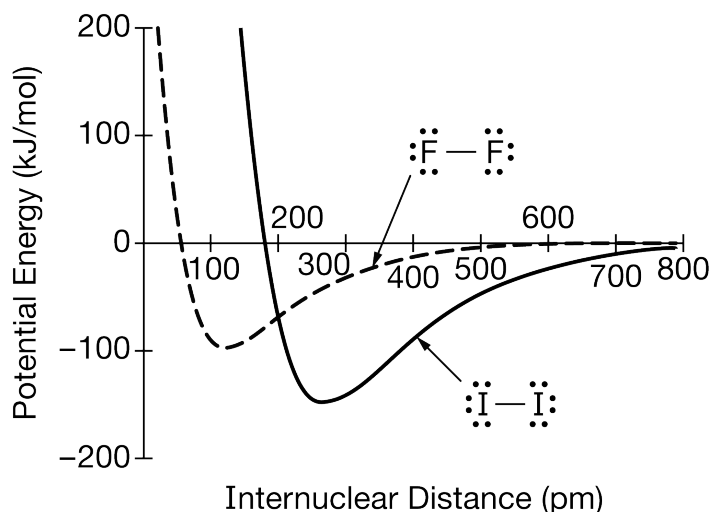
Fluorine can also react with another gas, $E(g)$, according to the following equation.



- (g) The student claims that the $D - F$ bond energy is greater than the $E - F$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two iodine atoms versus the distance between their nuclei. The student also finds that the $I - I$ bond energy is slightly less than the $F - F$ bond energy. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.

Chapter 09 Part 2: Rate Law



(h) Explain the error with the student's sketch for $\text{F}_2(g)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification, based on reaction order and initial rate data, equivalent to the following.

· Disagree. Comparing trials 1 and 3, when the initial pressure of F_2 is doubled as the initial pressure of D is held constant, the initial rate of reaction quadruples. This result is consistent only with a reaction that is second order with respect to F_2 .

· Response could also use mathematical justification.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct initial rate of F_2 and a valid justification equivalent to one of the following.

$$\frac{20.0 \times 10^{-3} \text{ torr DF}_4}{1 \text{ min}} \times \frac{2 \text{ torr F}_2}{1 \text{ torr DF}_4} = \frac{4.0 \times 10^{-2} \text{ torr F}_2}{1 \text{ min}}$$

Chapter 09 Part 2: Rate Law

OR

· The stoichiometric coefficient of F_2 in the chemical equation means that F_2 disappears at twice the rate that DF_4 appears, so the rate is $\frac{4.0 \times 10^{-2} \text{ torr } F_2}{1 \text{ min}}$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates agreement and provides a valid justification equivalent to the following.

· Agree. Since D is in group 18, the compound has 36 total valence electrons, which is consistent with a square planar geometry with four bonds and two lone pairs around the central atom.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct pressure.

· $12 \text{ atm} \times \frac{3 \text{ mol D}}{15 \text{ total mol gas}} = 2.4 \text{ atm}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct number of moles of each reactant.

· $D = 13 \text{ moles}$

· $F_2 = 22 \text{ moles}$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 2: Rate Law



0	1
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The response provides the correct energy and a valid justification equivalent to the following.

$$0.50 \text{ mol DF}_4 \times \frac{1 \text{ mol}_{rxn}}{1 \text{ mol DF}_4} \times \frac{-250 \text{ kJ}}{1 \text{ mol}_{rxn}} = -125 \text{ kJ} \approx -130 \text{ kJ}, \text{ so } 130 \text{ kJ is released.}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. Comparing the ΔH° values of the two reactions, the formation of DF_4 releases more energy than the formation of EF_4 . Therefore, the $\text{D} - \text{F}$ bond energy must be greater than the $\text{E} - \text{F}$ bond energy.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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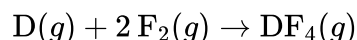
The response provides an explanation of the incorrect bond energy representation equivalent to the following.

· The minimum of the curve for F_2 should be lower than that of I_2 , to reflect that the $\text{F} - \text{F}$ bond is stronger than the $\text{I} - \text{I}$ bond.

Chapter 09 Part 2: Rate Law

12. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

An unknown gas, D, reacts with fluorine gas to form the compound $\text{DF}_4(g)$, as represented by the following equation.



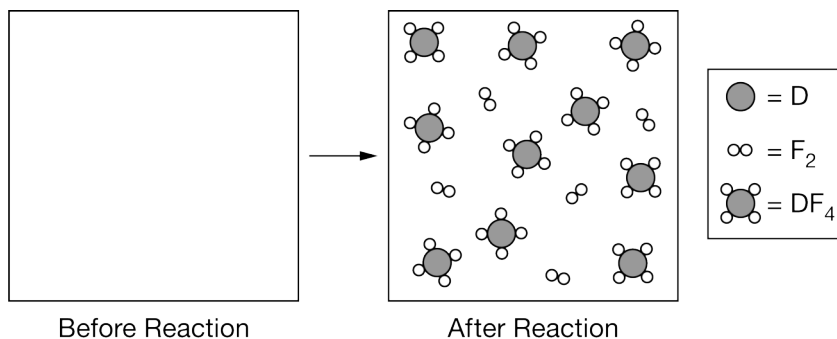
A series of experimental trials were performed at 600°C to determine the rate law expression for the reaction. Data from the trials are shown below.

	Initial P_{D} (torr)	Initial P_{F_2} (torr)	Initial Rate of Appearance of $\text{DF}_4(g)$ (torr/min)
Trial 1	3000	3000	2.50×10^{-2}
Trial 2	6000	3000	5.00×10^{-2}
Trial 3	3000	6000	10.0×10^{-2}

- (a) A student claims that the reaction is first order with respect to $\text{D}(g)$. Do you agree or disagree? Justify your answer using the data above.
- (b) What is the initial rate of disappearance of $\text{F}_2(g)$ in trial 1? Justify your answer.
- (c) A student claims that if element D is in group 18, then the molecular geometry of $\text{DF}_4(g)$ is tetrahedral. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\text{D}(g)$ and $\text{F}_2(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to “Before Reaction” is intentionally left blank.)

Chapter 09 Part 2: Rate Law

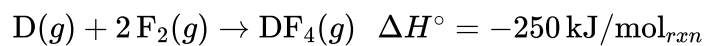


- (d) If the total pressure at the end of the reaction is 12 atm, what is the partial pressure of $\text{DF}_4(g)$?
- (e) Assuming that only D and F_2 are present in the container before the reaction, determine the number of moles of D and F_2 that were present in the initial reaction mixture.

D = _____

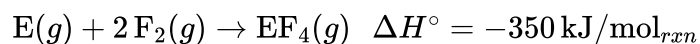
F_2 = _____

The equation and enthalpy change for the reaction between $\text{D}(g)$ and $\text{F}_2(g)$ are given below.



- (f) How much heat is released when 4.0 mol of $\text{F}_2(g)$ reacts with excess $\text{D}(g)$?

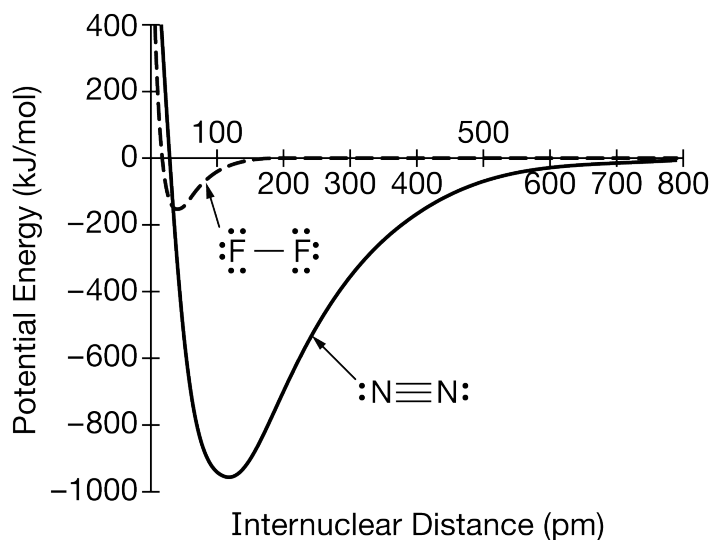
Fluorine can also react with another gas, $\text{E}(g)$, according to the following equation.



- (g) The student claims that the D – F bond energy is greater than the E – F bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two nitrogen atoms versus the distance between their nuclei. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.

Chapter 09 Part 2: Rate Law



(h) Explain the error with the student's sketch for $\text{F}_2(g)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification, based on reaction order and initial rate data, equivalent to the following.

- Agree. Comparing trials 1 and 2, when the initial pressure of D is doubled as the initial pressure of F_2 is held constant, the initial rate of reaction also doubles. This result is consistent only with a reaction that is first order with respect to D.
- Response could also use mathematical justification.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct initial rate of F_2 and a valid justification equivalent to one of the following.

$$\cdot \frac{2.50 \times 10^{-2} \text{ torr DF}_4}{1 \text{ min}} \times \frac{2 \text{ torr F}_2}{1 \text{ torr DF}_4} = \frac{5.00 \times 10^{-2} \text{ torr F}_2}{1 \text{ min}}$$

Chapter 09 Part 2: Rate Law

OR

· The stoichiometric coefficient of F_2 in the chemical equation means that F_2 disappears at twice the rate that DF_4 appears, so the rate is $\frac{5.00 \times 10^{-2} \text{ torr } F_2}{1 \text{ min}}$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. Since D is in group 18, the compound has 36 total valence electrons, which is not consistent with a tetrahedral geometry. Rather, the geometry would be square planar with four bonds and two lone pairs around the central atom.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct pressure.

· $12 \text{ atm} \times \frac{10 \text{ mol } DF_4}{15 \text{ total mol gas}} = 8.0 \text{ atm}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct number of moles of each reactant.

· $D = 10 \text{ moles}$

· $F_2 = 25 \text{ moles}$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 2: Rate Law



0	1
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The response provides the correct energy and a valid justification equivalent to the following.

$$4.0 \text{ mol F}_2 \times \frac{1 \text{ mol}_{rxn}}{2 \text{ mol F}_2} \times \frac{-250 \text{ kJ}}{1 \text{ mol}_{rxn}} = -500 \text{ kJ, so 500 kJ is released}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

Disagree. Comparing the ΔH° values of the two reactions, the formation of DF_4 releases less energy than the formation of EF_4 . Therefore, the $\text{D} - \text{F}$ bond energy must be less than the $\text{E} - \text{F}$ bond energy.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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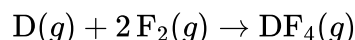
The response provides an explanation of the incorrect bond length representation equivalent to the following.

The minimum of the curve for F_2 should be to the right of that for N_2 to reflect that the $\text{F} - \text{F}$ single bond is longer than the $\text{N} \equiv \text{N}$.

Chapter 09 Part 2: Rate Law

13. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

An unknown gas, D, reacts with fluorine gas to form the compound $\text{DF}_4(g)$, as represented by the following equation.



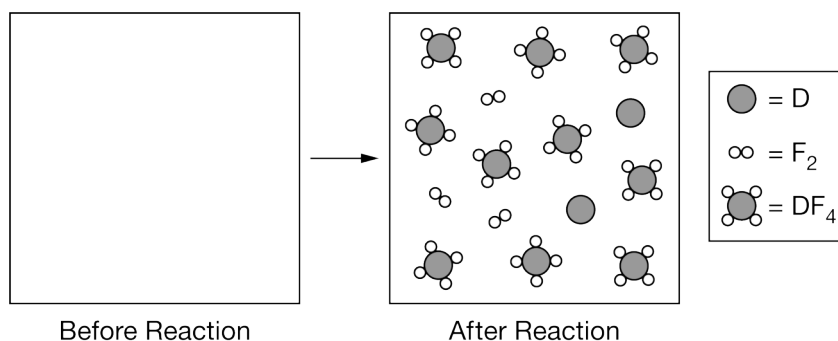
A series of experimental trials were performed at 600°C to determine the rate law expression for the reaction. Data from the trials are shown below.

	Initial P_{D} (torr)	Initial P_{F_2} (torr)	Initial Rate of Appearance of $\text{DF}_4(g)$ (torr/min)
Trial 1	3000	3000	2.50×10^{-2}
Trial 2	6000	3000	5.00×10^{-2}
Trial 3	3000	6000	10.0×10^{-2}

- (a) A student claims that the reaction is second order with respect to $\text{F}_2(g)$. Do you agree or disagree? Justify your answer using the data above.
- (b) What is the initial rate of disappearance of $\text{D}(g)$ in trial 3? Justify your answer.
- (c) A student claims that if element D is in group 16, then the molecular geometry of $\text{DF}_4(g)$ is see-saw. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\text{D}(g)$ and $\text{F}_2(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to “Before Reaction” is intentionally left blank.)

Chapter 09 Part 2: Rate Law

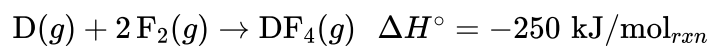


- (d) If the total pressure at the end of the reaction is 12 atm, what is the partial pressure of D(*g*)?
- (e) Assuming that only D and F₂ are present in the container before the reaction, determine the number of moles of D and F₂ that were present in the initial reaction mixture.

D = _____

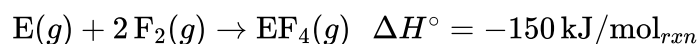
F₂ = _____

The equation and enthalpy change for the reaction between D(*g*) and F₂(*g*) are given below.



- (f) How much heat is released when 0.75 mol of D(*g*) reacts with excess F₂(*g*) ?

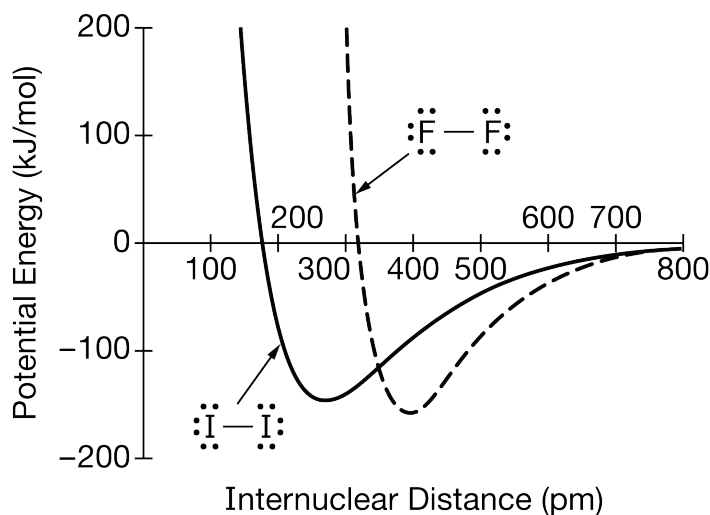
Fluorine can also react with another gas, E(*g*), according to the following equation.



- (g) The student claims that the D – F bond energy is less than the E – F bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the following diagram, which shows the potential energy of two iodine atoms versus the distance between their nuclei. The student also finds that the I – I bond energy is slightly less than the F – F bond energy. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.

Chapter 09 Part 2: Rate Law



(h) Explain the error with the student's sketch for $\text{F}_2(g)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification, based on reaction order and initial rate data, equivalent to the following.

- Agree. Comparing trials 1 and 3, when the initial pressure of F_2 is doubled as the initial pressure of D is held constant, the initial rate of reaction quadruples. This result is consistent only with a reaction that is second order with respect to F_2 .
- Response could also use mathematical justification.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct initial rate of D and a valid justification equivalent to one of the following.

$$\frac{10.00 \times 10^{-2} \text{ torr DF}_4}{1 \text{ min}} \times \frac{1 \text{ torr D}}{1 \text{ torr DF}_4} = \frac{10.00 \times 10^{-2} \text{ torr D}}{1 \text{ min}}$$

Chapter 09 Part 2: Rate Law

OR

· The stoichiometric coefficient of D in the chemical equation means that D disappears at the same rate that DF_4 appears, so the rate is $\frac{10.00 \times 10^{-2} \text{ torr D}}{1 \text{ min}}$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. Since D is in group 16, the compound has 34 total valence electrons, which is consistent with a see-saw geometry with four bonds and one lone pair around the central atom.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct pressure.

· $12 \text{ atm} \times \frac{2 \text{ mol D}}{15 \text{ total mol gas}} = 1.6 \text{ atm}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct number of moles of each reactant.

· $\text{D} = 12 \text{ moles}$

· $\text{F}_2 = 23 \text{ moles}$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 2: Rate Law



0	1
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The response provides the correct energy and a valid justification equivalent to the following.

$$0.75 \text{ mol D} \times \frac{1 \text{ mol}_{\text{rxn}}}{1 \text{ mol D}} \times \frac{-250 \text{ kJ}}{1 \text{ mol}_{\text{rxn}}} = -187.5 \text{ kJ} \approx -190 \text{ kJ}, \text{ so } 190 \text{ kJ is released}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

Disagree. Comparing the ΔH° values of the two reactions, the formation of DF_4 releases more energy than the formation of EF_4 . Therefore, the $\text{D} - \text{F}$ bond energy must be greater than the $\text{E} - \text{F}$ bond energy.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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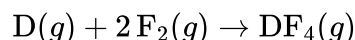
The response provides an explanation of the incorrect bond length representation equivalent to the following.

The minimum of the curve for F_2 should be to the left of that for I_2 . The atomic radius of F is smaller than the atomic radius of I, leading to the $\text{F} - \text{F}$ bond being shorter than the $\text{I} - \text{I}$ bond.

Chapter 09 Part 2: Rate Law

14. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

An unknown gas, D, reacts with fluorine gas to form the compound $\text{DF}_4(g)$, as represented by the following equation.



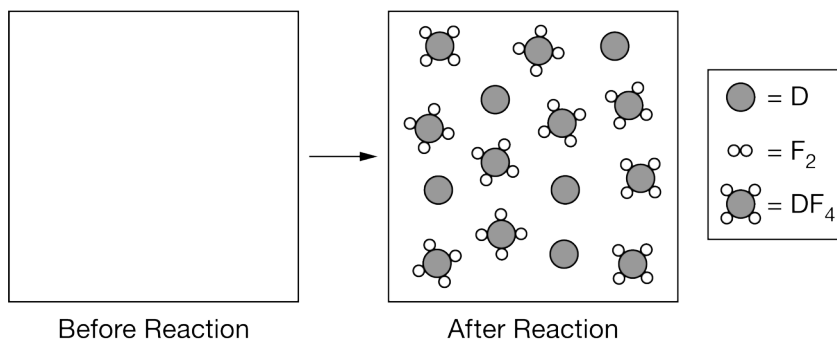
A series of experimental trials were performed at 600°C to determine the rate law expression for the reaction. Data from the trials are shown below.

	Initial P_{D} (torr)	Initial P_{F_2} (torr)	Initial Rate of Appearance of $\text{DF}_4(g)$ (torr/min)
Trial 1	3000	3000	2.50×10^{-2}
Trial 2	6000	3000	5.00×10^{-2}
Trial 3	3000	6000	10.0×10^{-2}

- (a) A student claims that the reaction is second order with respect to $\text{D}(g)$. Do you agree or disagree? Justify your answer using the data above.
- (b) What is the initial rate of disappearance of $\text{F}_2(g)$ in trial 2? Justify your answer.
- (c) A student claims that if element D is in group 16, then the molecular geometry of $\text{DF}_4(g)$ is tetrahedral. Do you agree or disagree with the student? Justify your answer in terms of VSEPR theory.

The particulate model shown below represents the moles of particles present in the container after $\text{D}(g)$ and $\text{F}_2(g)$ have reacted in a different experiment. Assume that each particle in the diagram represents one mole of the respective substance. (Note: The box corresponding to “Before Reaction” is intentionally left blank.)

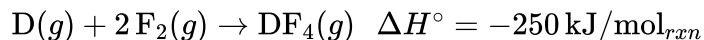
Chapter 09 Part 2: Rate Law



- (d) If the total pressure at the end of the reaction is 12 atm, what is the partial pressure of $D(g)$?
- (e) Assuming that only D and F_2 are present in the container before the reaction, determine the number of moles of D and F_2 that were present in the initial reaction mixture.

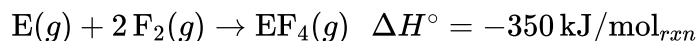
Math output error

The equation and enthalpy change for the reaction between $D(g)$ and $F_2(g)$ are given below.



- (f) How much heat is released when 0.50 mol of $F_2(g)$ reacts with excess $D(g)$?

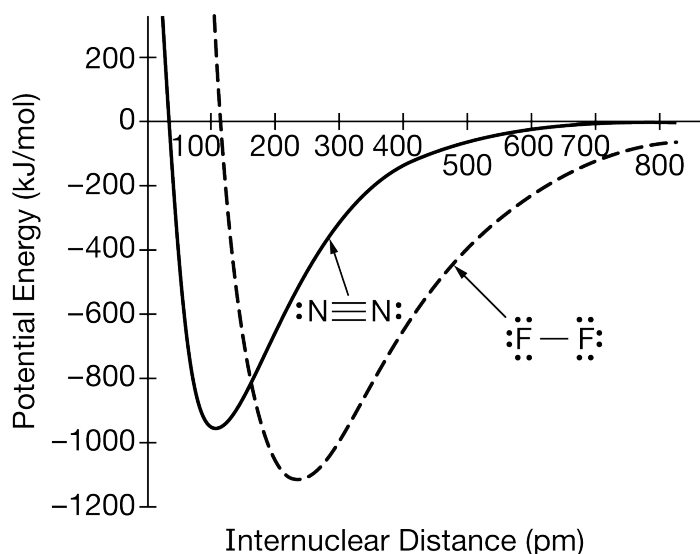
Fluorine can also react with another gas, $E(g)$, according to the following equation.



- (g) The student claims that the $D - F$ bond energy is less than the $E - F$ bond energy. Do you agree or disagree? Justify your answer.

While investigating the properties of other diatomic molecules, the student finds the diagram below, which shows the potential energy of two nitrogen atoms versus the distance between their nuclei. The student incorrectly sketches the potential energy curve for two fluorine atoms, shown by the dotted line in the diagram.

Chapter 09 Part 2: Rate Law



(h) Explain the error with the student's sketch for $F_2(g)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification, based on reaction order and initial rate data, equivalent to the following.

- Disagree. Comparing trials 1 and 2, when the initial pressure of D is doubled as the initial pressure of F_2 is held constant, the initial rate of reaction also doubles. This result is consistent only with a reaction that is first order with respect to D.
- Response could also use mathematical justification.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct initial rate of F_2 and a valid justification equivalent to one of the following.

Chapter 09 Part 2: Rate Law

$$\cdot \frac{5.00 \times 10^{-2} \text{ torr DF}_4}{1 \text{ min}} \times \frac{2 \text{ torr F}_2}{1 \text{ torr DF}_4} = \frac{1.00 \times 10^{-1} \text{ torr F}_2}{1 \text{ min}}$$

OR

· The stoichiometric coefficient of F_2 in the chemical equation means that F_2 disappears at twice the rate that DF_4 appears, so the rate is $\frac{1.00 \times 10^{-1} \text{ torr F}_2}{1 \text{ min}}$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. Since D is in group 16, the compound has 34 total valence electrons, which is not consistent with a tetrahedral geometry. Rather, the geometry would be see-saw with four bonds and one lone pair around the central atom.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct pressure.

$$\cdot 12 \text{ atm} \times \frac{5 \text{ mol D}}{15 \text{ total mol gas}} = 4.0 \text{ atm}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct number of moles of each reactant.

· D = 15 moles

· F_2 = 20 moles

Part F

Chapter 09 Part 2: Rate Law

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct energy and a valid justification equivalent to the following.

$$0.50 \text{ mol F}_2 \times \frac{1 \text{ mol}_{rxn}}{2 \text{ mol F}_2} \times \frac{-250 \text{ kJ}}{1 \text{ mol}_{rxn}} = -62.5 \text{ kJ} \approx -63 \text{ kJ}, \text{ so } 63 \text{ kJ is released}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

Disagree. Comparing the ΔH° values of the two reactions, the formation of DF_4 releases less energy than the formation of EF_4 . Therefore, the D — F bond energy must be less than the E — F bond energy.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides an explanation of the incorrect bond energy representation equivalent to the following.

The minimum of the curve for F_2 should be higher than that of N_2 , to reflect that the F — F single bond is weaker than the $\text{N} \equiv \text{N}$.

15. If the oxygen isotope ^{20}O has a half-life of 15 seconds, what fraction of a sample of pure ^{20}O remains after 1.0 minute?

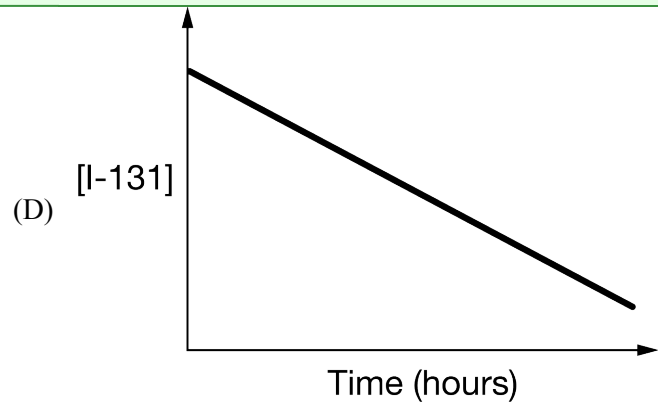
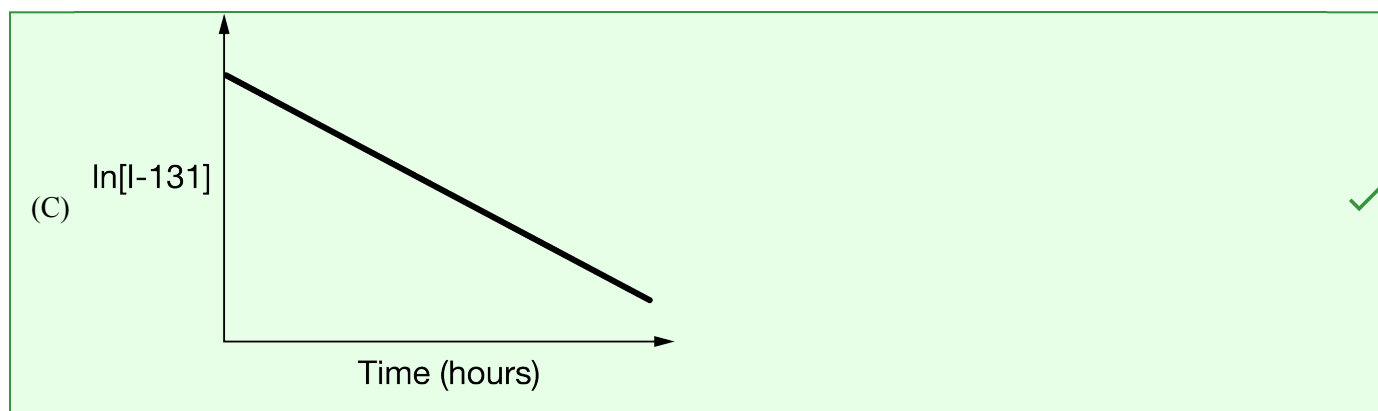
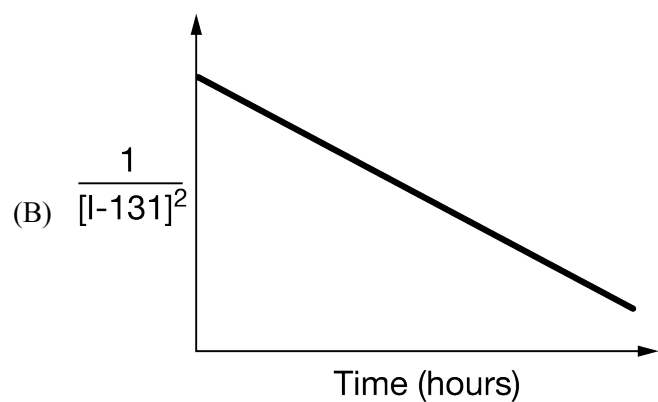
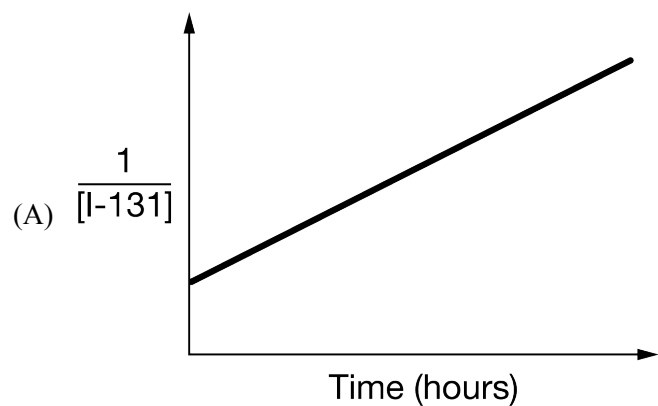
- (A) $\frac{1}{2}$
(B) $\frac{1}{4}$
(C) $\frac{7}{30}$
(D) $\frac{1}{8}$
(E) $\frac{1}{16}$



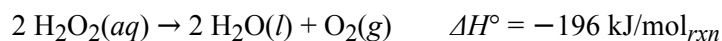
Chapter 09 Part 2: Rate Law

16. Gaseous cyclobutene undergoes a first-order reaction to form gaseous butadiene. At a particular temperature, the partial pressure of cyclobutene in the reaction vessel drops to one-eighth its original value in 124 seconds. What is the half-life for this reaction at this temperature?
- (A) 15.5 sec
(B) 31.0 sec
(C) 41.3 sec ✓
(D) 62.0 sec
(E) 124 sec
17. The isomerization of cyclopropane to propylene is a first-order process with a half-life of 19 minutes at 500°C. The time it takes for the partial pressure of cyclopropane to decrease from 1.0 atmosphere to 0.125 atmosphere at 500°C is closest to
- (A) 38 minutes
(B) 57 minutes ✓
(C) 76 minutes
(D) 152 minutes
(E) 190 minutes
18. The rate constant (k) for the decay of the radioactive isotope I-131 is $3.6 \times 10^{-3} \text{ hours}^{-1}$. The slope of which of the following graphs is correct for the decay and could be used to confirm the value of k ?

Chapter 09 Part 2: Rate Law



Chapter 09 Part 2: Rate Law



The decomposition of $\text{H}_2\text{O}_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $\text{H}_2\text{O}_2(aq)$ at a constant temperature of 300. K and recorded the concentration of H_2O_2 as a function of time. The results are given in the table below.

Time (s)	$[\text{H}_2\text{O}_2]$
0	2.7
200.	2.1
400.	1.7
600.	1.4

19. Which of the following statements is a correct interpretation of the data regarding how the order of the reaction can be determined?

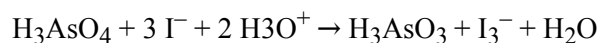
(A) The reaction must be first order because there is only one reactant species.

(B) The reaction is first order if the plot of $\ln [\text{H}_2\text{O}_2]$ versus time is a straight line. ✓

(C) The reaction is first order if the plot of $1/[\text{H}_2\text{O}_2]$ versus time is a straight line.

(D) The reaction is second order because 2 is the coefficient of H_2O_2 in the chemical equation.

20.



The oxidation of iodide ions by arsenic acid in acidic aqueous solution occurs according to the stoichiometry shown above. The experimental rate law of the reaction is:

$$\text{Rate} = k[\text{H}_3\text{AsO}_4] [\text{I}^-] [\text{H}_3\text{O}^+]$$

What is the order of the reaction with respect to I^- ?

(A) 1 ✓

(B) 2

(C) 3

(D) 5

(E) 6

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21. If 87.5 percent of a sample of pure ^{131}I decays in 24 days, what is the half-life of ^{131}I ?

(A) 6 days

(B) 8 days

(C) 12 days

(D) 14 days

(E) 21 days

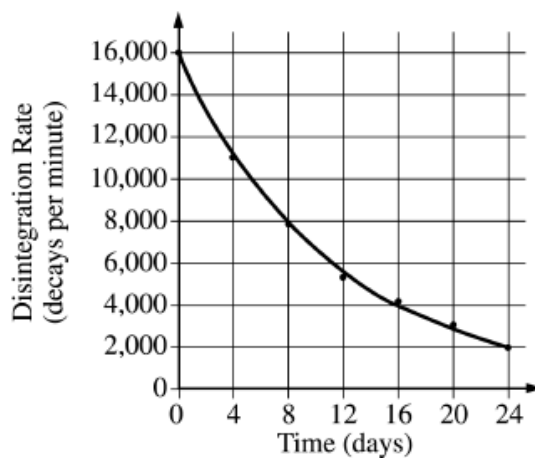


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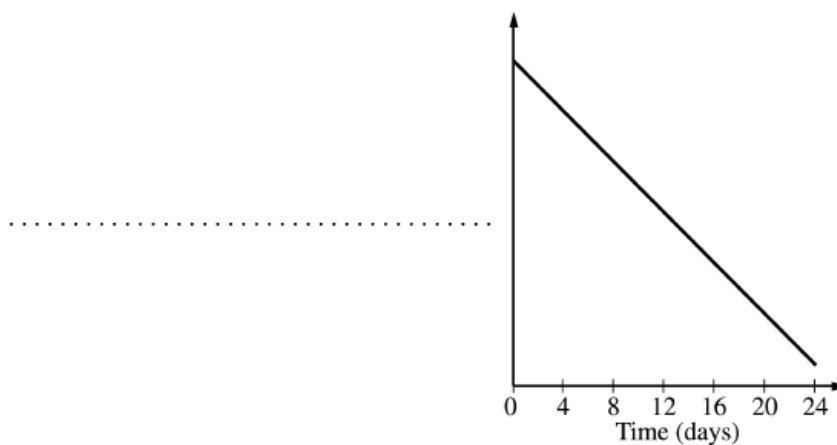
22. The decay of the radioisotope I-131 was studied in a laboratory. I-131 is known to decay by beta (${}_{-1}^0e$) emission.

- Write a balanced nuclear equation for the decay of I-131.
- What is the source of the beta particle emitted from the nucleus?

The radioactivity of a sample of I-131 was measured. The data collected are plotted on the graph below.



- Determine the half-life, $t_{1/2}$, of I-131 using the graph above.
- The data can be used to show that the decay of I-131 is a first-order reaction, as indicated on the graph below.



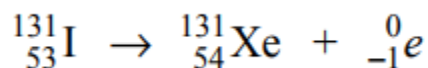
- Label the vertical axis of the graph above.
- What are the units of the rate constant, k , for the decay reaction?

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- iii. Explain how the half-life of I-131 can be calculated using the slope of the line plotted on the graph.
- e. Compare the value of the half-life of I-131 at 25°C to its value at 50°C.

Part A

1 point for correct equation

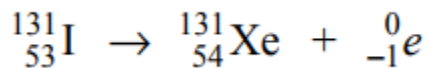


✓

0	1
---	---

The student response earns one of the following points:

1 point for correct equation

**Part B**

1 point for identifying a neutron as the source of the beta emission

A neutron spontaneously decays to an electron and a proton.

✓

0	1
---	---

The student response earns one of the following points:

1 point for identifying a neutron as the source of the beta emission

A neutron spontaneously decays to an electron and a proton.

Part C

1 point for half-life

The half-life is 8 days. That is the time required for the disintegration rate to fall from 16,000 to one-half its initial value, 8,000.

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0	1
---	---

The student response earns one of the following points:

1 point for half-life

The half-life is 8 days. That is the time required for the disintegration rate to fall from 16,000 to one-half its initial value, 8,000.

Part C (i,ii,iii)

(i)

1 point for correct label on y-axis

The label on the y-axis should be ln or log one of the following: disintegrations or moles or atoms or [I-131] or disintegration rate.

(ii)

1 point for correct units

From the graph, the units on the rate constant are days⁻¹ (Units of time⁻¹ is acceptable)

(iii)

1 point for indicating slope is k

1 point for half-life equation

The slope of the line is $-k$. The slope is negative, so k is a positive number. The half-life can then be calculated using the relationship $t_{1/2} =$

$$\frac{0.693}{k}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

(i)

1 point for correct label on y-axis

The label on the y-axis should be ln or log one of the following: disintegrations or moles or atoms or [I-131] or

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disintegration rate.

(ii)

1 point for correct units

From the graph, the units on the rate constant are days^{−1} (Units of time^{−1} is acceptable)

(iii)

1 point for indicating slope is k

1 point for half-life equation

The slope of the line is $-k$. The slope is negative, so k is a positive number. The half-life can then be calculated using the relationship $t_{1/2} =$

$$\frac{0.693}{k}$$

Part D

1 point for answer

The half-life will be the same at the different temperatures. The half-life of a nuclear decay process is independent of temperature.



0	1
---	---

The student response earns one of the following points:

1 point for answer

The half-life will be the same at the different temperatures. The half-life of a nuclear decay process is independent of temperature.

23. The half-life of ^{55}Cr is about 2.0 hours. The delivery of a sample of this isotope from the reactor to a certain laboratory requires 12 hours. About what mass of such material should be shipped in order that 1.0 mg of ^{55}Cr is delivered to the laboratory?

(A) 130 mg

(B) 64 mg

(C) 32 mg

(D) 11 mg

(E) 1.0 mg

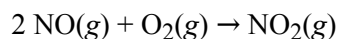


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24.

Experiment	Initial [NO] (mol L ⁻¹)	Initial [O ₂] (mol L ⁻¹)	Initial Rate of Formation of NO ₂ (mol L ⁻¹ s ⁻¹)
1	0.10	0.10	2.5×10^{-4}
2	0.20	0.10	5.0×10^{-4}
3	0.20	0.40	8.0×10^{-3}

The initial-rate data in the table above were obtained for the reaction represented below. What is the experimental rate law for the reaction?



(A) $\text{Rate} = k[\text{NO}][\text{O}_2]$

(B) $\text{Rate} = k[\text{NO}]^2[\text{O}_2]$

(C) $\text{Rate} = k[\text{NO}]^2[\text{O}_2]$

(D) $\text{Rate} = k[\text{NO}]^2[\text{O}_2]^2$

(E) $\text{Rate} = k[\text{NO}] / [\text{O}_2]$

25.

Experiment	[X] ₀	[Y] ₀	Initial Rate of Formation of Z (mol L ⁻¹ sec ⁻¹)
1	0.40	0.10	R
2	0.20	0.20	?

The table above shows the results from a rate study of the reaction $\text{X} + \text{Y} \rightarrow \text{Z}$. Starting with known concentrations of X and Y in experiment 1, the rate of formation of Z was measured. If the reaction was first order with respect to X and second order with respect to Y, the initial rate of formation of Z in experiment 2 would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) R

(D) $2R$

(E) $4R$

26.

$$\text{Rate} = k[\text{M}][\text{N}]^2$$

The rate of a certain chemical reaction between substances M and N obeys the rate law above. The reaction is first studied with [M] and [N] each 1×10^{-3} molar. If a new experiment is conducted with [M] and [N] each 2×10^{-3} molar, the reaction rate will increase by a factor of

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(A) 2

(B) 4

(C) 6

(D) 8

(E) 16



27.

Trial	$[A_2]$	$[B]$	Initial rate ($M s^{-1}$)
1	0.10	0.50	2.5×10^{-4}
2	0.20	0.50	5.0×10^{-4}
3	0.30	0.05	5.0×10^{-5}
4	0.30	0.10	1.0×10^{-4}

An experiment was conducted to determine the rate law for the reaction $A_2(g) + B(g) \rightarrow A_2B(g)$. The table above shows the data collected. Based on the data in the table, which statement is correct?

(A) Since the rate law can be expressed as $rate = k[A_2]^2$, tripling the concentration of A_2 will cause a 9-fold increase in the rate of the reaction.

(B) Since the rate law can be expressed as $rate = k[A_2][B]$, doubling the concentrations of A_2 and B will quadruple the rate of the reaction.



(C) Since the rate law can be expressed as $rate = k[A_2]^2[B]$, tripling the concentration of A_2 while keeping the concentration of B constant will triple the rate of the reaction.

(D) Since the rate law can be expressed as $rate = k[A_2][B]^2$, doubling the concentration of B while keeping the concentration of A_2 constant will double the rate of the reaction.

28.

Time (s)	$[NO_2]$ (mol/L)	$\ln [NO_2]$	$\frac{1}{[NO_2]}$ (L/mol)
0	0.500	-0.693	2.00
100	0.364	-1.01	2.75
200	0.286	-1.25	3.50
300	0.235	-1.45	4.25

The data from a study of the decomposition of $NO_2(g)$ to form $NO(g)$ and $O_2(g)$ are given in the table above. Which of the following rate laws is consistent with the data?

(A) $Rate = k[NO_2]$

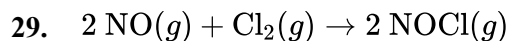
(B) $Rate = k[NO_2]^2$



(C) $Rate = k \frac{1}{[NO_2]}$

(D) $Rate = k \frac{1}{[NO_2]^2}$

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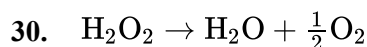


Experiment	[NO]	[Cl ₂]	Initial Rate of NOCl Formation (M s ⁻¹)
1	0.0250	0.0510	4.55×10^{-5}
2	0.0250	0.1020	9.10×10^{-5}
3	0.0500	0.0510	1.82×10^{-4}

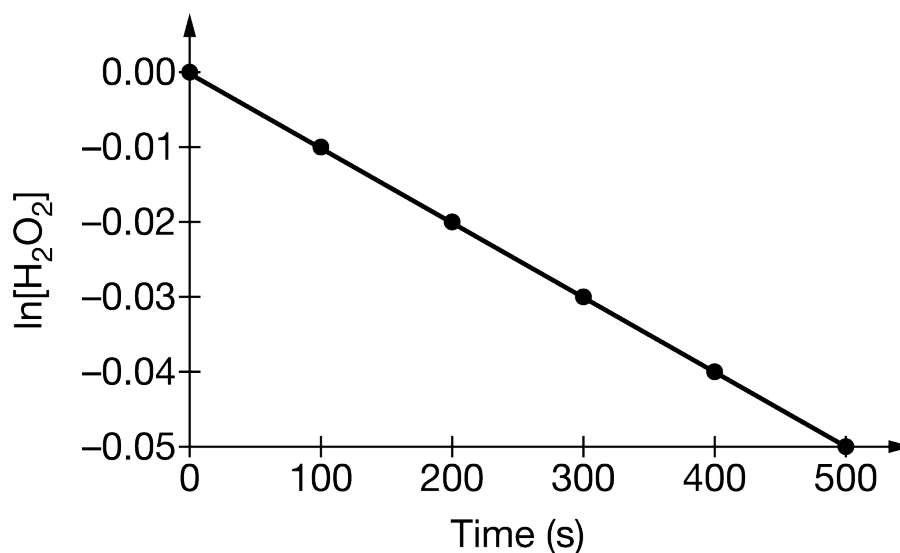
The initial rates of the reaction represented by the equation shown above were measured for different initial concentrations of $\text{NO}(g)$ and $\text{Cl}_2(g)$. Based on the data given in the table above, which of the following is the rate law expression for the reaction, and why?

- (A) $\text{Rate} = k[\text{NO}]^2$, because the initial rate quadrupled when [NO] was doubled but remained constant when [Cl₂] was doubled.
- (B) $\text{Rate} = k[\text{NO}][\text{Cl}_2]$, because the initial rate doubled when either [NO] or [Cl₂] was doubled.
- (C) $\text{Rate} = k[\text{NO}][\text{Cl}_2]^2$, because the initial rate doubled when [NO] was doubled and quadrupled when [Cl₂] was doubled.
- (D) $\text{Rate} = k[\text{NO}]^2[\text{Cl}_2]$, because the initial rate quadrupled when [NO] was doubled and doubled when [Cl₂] was doubled. ✓

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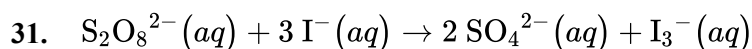


Hydrogen peroxide decomposes to produce water and oxygen according to the equation above. An experimentally determined graph for the first-order decomposition of hydrogen peroxide is provided below.



Which of the following best identifies the rate constant k for the reaction based on the information in the plot of $\ln[\text{H}_2\text{O}_2]$ versus time (t) ?

- (A) $k = \ln[\text{H}_2\text{O}_2]$ at $t = 0$ s
- (B) $k = \ln[\text{H}_2\text{O}_2]$ at $t = 500$ s
- (C) $k = -(\text{slope of plot})$
- (D) $k = -\frac{1}{(\text{slope of plot})}$



In aqueous solution, the reaction represented by the balanced equation shown above has the experimentally determined rate law: $\text{rate} = k[\text{S}_2\text{O}_8^{2-}][\text{I}^-]$.

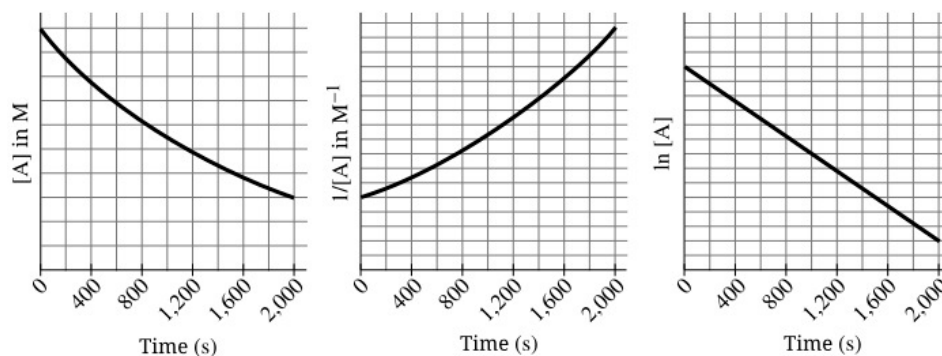
If the concentration of $[\text{S}_2\text{O}_8^{2-}]$ is doubled while keeping $[\text{I}^-]$ constant, which of the following experimental results is predicted based on the rate law, and why?

- (A) The rate of reaction will remain the same, because k will decrease by half.
- (B) The rate of reaction will double, because the rate is directly proportional to $[\text{S}_2\text{O}_8^{2-}]$.
- (C) The rate of reaction will increase by a factor of four, because two moles of SO_4^{2-} are produced for each mole of $\text{S}_2\text{O}_8^{2-}$ consumed.
- (D) The rate of reaction will increase by a factor of four, because the reaction is second order overall.



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32.



A student recorded the rate of decomposition of compound **A** as a function of time at 45 °C. The experimental data are shown above. What is the rate law for the decomposition of **A** at this temperature?

(A) $\text{Rate} = k$

(B) $\text{Rate} = k[\text{A}]$

(C) $\text{Rate} = k[\text{A}]^2$

(D) The rate law cannot be determined without additional experimentation.

33. The rate law for the reaction of nitrogen dioxide and chlorine is found to be $\text{rate} = k [\text{NO}_2]^2 [\text{Cl}_2]$. By what factor does the rate of the reaction change when the concentrations of both NO_2 and Cl_2 are doubled?

(A) 2

(B) 3

(C) 4

(D) 6

(E) 8

34.

Experiment	Initial [X] (mol L ⁻¹)	Initial [Y] (mol L ⁻¹)	Initial Rate of Formation of Z (mol L ⁻¹ s ⁻¹)
1	0.10	0.30	4.0×10^{-4}
2	0.20	0.60	1.6×10^{-3}
3	0.20	0.30	4.0×10^{-4}

The data in the table above were obtained for the reaction $X + Y \rightarrow Z$. Which of the following is the rate law for the reaction?

(A) $\text{Rate} = k[\text{X}]^2$

(B) $\text{Rate} = k[\text{Y}]^2$

(C) $\text{Rate} = k[\text{X}][\text{Y}]$

(D) $\text{Rate} = k[\text{X}]^2[\text{Y}]$

(E) $\text{Rate} = k[\text{X}][\text{Y}]^2$

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35.

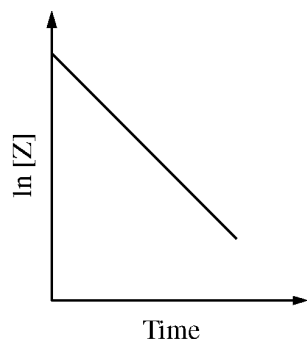
Time (days)	0	1	2	3	4	5	6	7	...	10	...	20
% Reactant remaining	100	79	63	50	40	31	25	20		10		1

A reaction was observed for 20 days and the percentage of the reactant remaining after each day was recorded in the table above. Which of the following best describes the order and the half-life of the reaction?

(A)	<u>Reaction Order</u>	<u>Half-life (days)</u>	✓
	First	3	
(B)	<u>Reaction Order</u>	<u>Half-life (days)</u>	
	First	10	
(C)	<u>Reaction Order</u>	<u>Half-life (days)</u>	
	Second	3	
(D)	<u>Reaction Order</u>	<u>Half-life (days)</u>	
	Second	3	
(E)	<u>Reaction Order</u>	<u>Half-life (days)</u>	
	Second	10	

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36.



Consider the reaction represented by the equation $2 X + 2 Z \rightarrow X_2Z_2$. During a reaction in which a large excess of reactant X was present, the concentration of reactant Z was monitored over time. A plot of the natural logarithm of the concentration of Z versus time is shown in the figure above. The order of the reaction with respect to reactant Z is

(A) zero order

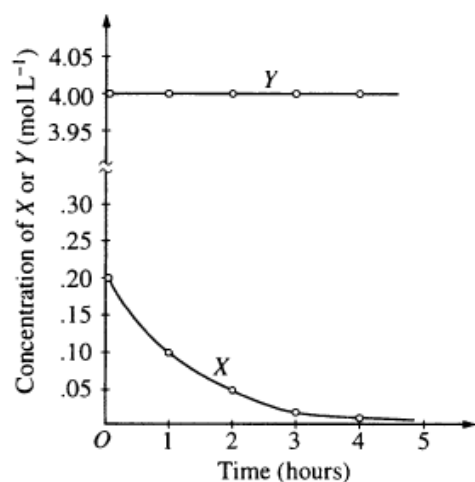
(B) first order

(C) second order

(D) third order



37.



The graph above shows the results of a study of the reaction of X with a large excess of Y to yield Z . The concentrations of X and Y were measured over a period of time. According to the results, which of the following can be concluded about the rate law for the reaction under the conditions studied?

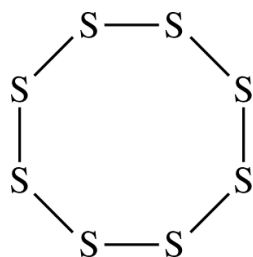
(A) It is zero order in $[X]$.(B) It is first order in $[X]$.(C) It is second order in $[X]$.(D) It is first order in $[Y]$.

(E) The overall order of the reaction is 2.



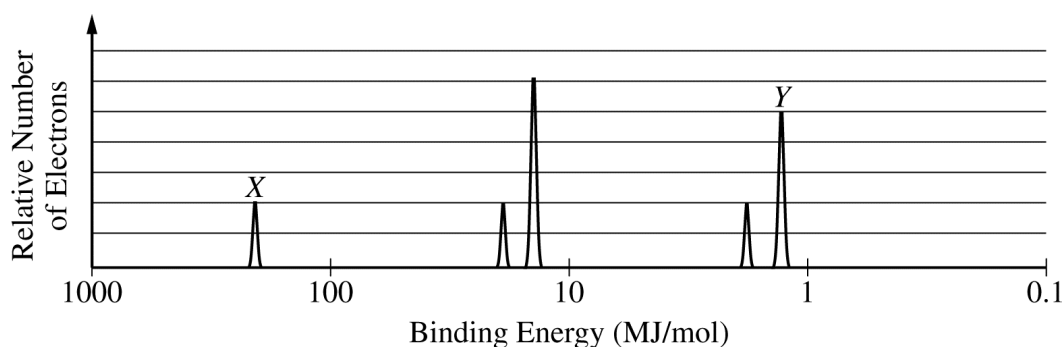
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38.

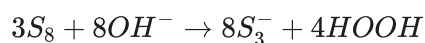


Elemental sulfur can exist as molecules with the formula S_8 . The S_8 molecule is represented by the incomplete Lewis diagram above.

- The diagram of S_8 shows only bonding pairs of electrons. How many lone pairs of electrons does each S atom in the molecule have?
- Based on your answer to part (a), determine the expected value of the S–S–S bond angles in the S_8 molecule.
- Write the electron configuration for the S atom in its ground state.
- The complete photoelectron spectrum for the element chlorine is represented below. Peak X in the spectrum corresponds to the binding energy of electrons in a certain orbital of chlorine atoms. The electrons in this orbital of chlorine have a binding energy of 273 MJ/mol, while the electrons in the same orbital of sulfur atoms have a binding energy of 239 MJ/mol



- Identify the orbital and explain the difference between the binding energies in terms of Coulombic forces.
- Peak Y corresponds to the electrons in certain orbitals of chlorine atoms. On the spectrum shown, carefully draw the peak that would correspond to the electrons in the same orbitals of sulfur atoms.



In an experiment, a student studies the kinetics of the reaction represented above and obtains the data

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shown in the following table.

Experiment	Initial $[S_8]$ (M)	Initial $[OH^-]$ (M)	Initial Reaction Rate (M/s)
Trial 1	0.100	0.0100	0.699
Trial 2	0.300	0.0100	2.10
Trial 3	0.300	?	4.19

- e. Use the data in the table to do the following.
- Determine the order of the reaction with respect to S_8 . Justify your answer.
 - Determine the value of $[OH^-]$ that was used in trial 3, considering that the reaction is first order with respect to OH^- . Justify your answer.

The next day the student conducts trial 4 using the same concentrations of S_8 and OH^- as in trial 1, but the reaction occurs at a much slower rate than the reaction in trial 1. The student observes that the temperature in the lab is lower than it was the day before.

- f. Using particle-level reasoning, provide TWO explanations that help to account for the fact that the reaction rate is slower in trial 4.

Part A

1 point is earned for the correct answer.

Two



0	1
---	---

Student earns 1/1 point

1 point is earned for the correct answer.

Two

Part B

1 point is earned for an angle in the acceptable range.

Chapter 09 Part 2: Rate Law

109.5°
 Acceptable range: $104^\circ \leq \text{angle} \leq 110^\circ$.
 (The experimentally determined angle is 107.8°)



0	1
---	---

Student earns 1/1 point

1 point is earned for an angle in the acceptable range.

109.5°
 Acceptable range: $104^\circ \leq \text{angle} \leq 110^\circ$.
 (The experimentally determined angle is 107.8°)

Part C

1 point is earned for a correct configuration.

1s22s22p63s23p4 OR [Ne] 3s23p4



0	1
---	---

Student earns 1/1 point

1 point is earned for a correct configuration.

1s22s22p63s23p4 OR [Ne] 3s23p4

Part D

Part i

1 point is earned for identifying the orbital as 1s.

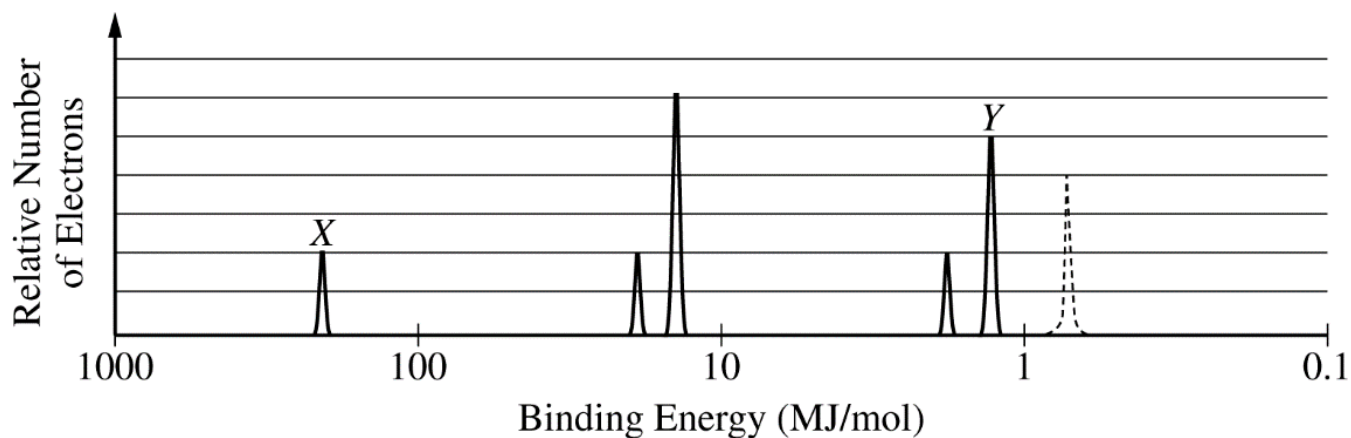
1 point is earned for a valid explanation

Peak *X* represents electrons in a 1s orbital. A Cl atom has one more proton in its nucleus than does a S atom; therefore, the electrons in Cl are more strongly attracted to the nucleus, and the binding energy of the 1s electrons in the Cl atom is greater than that of the 1s electrons in the S atom.

Part ii

1 point is earned for a peak that is to the right of peak *Y* and that has the correct height.

Chapter 09 Part 2: Rate Law



See example of a correct response (dashed peak) above.



0	1	2	3
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Student earns 3/3 point(s)

Part i

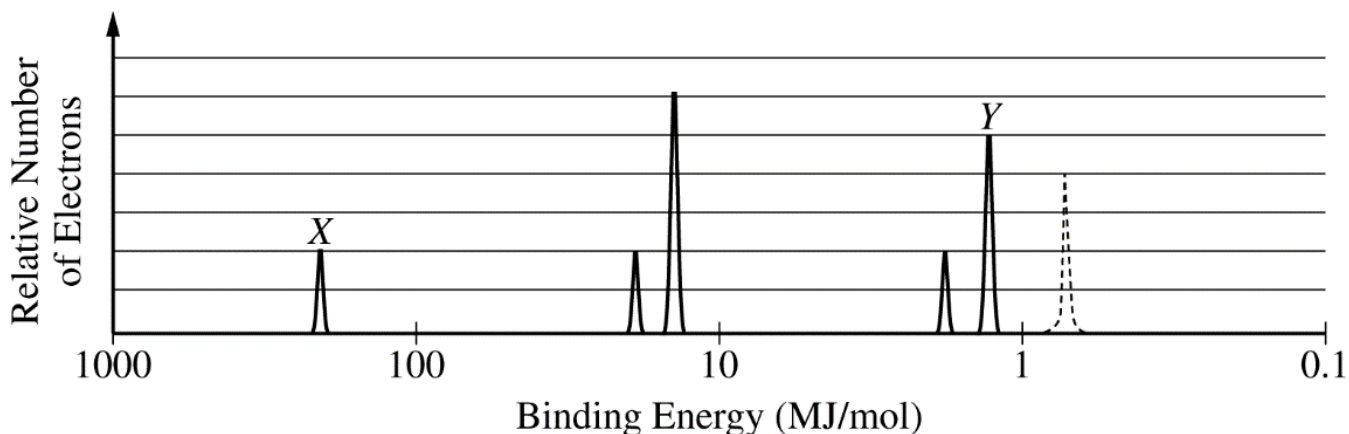
1 point is earned for identifying the orbital as 1s.

1 point is earned for a valid explanation

Peak *X* represents electrons in a 1s orbital. A Cl atom has one more proton in its nucleus than does a S atom; therefore, the electrons in Cl are more strongly attracted to the nucleus, and the binding energy of the 1s electrons in the Cl atom is greater than that of the 1s electrons in the S atom.

Part ii

1 point is earned for a peak that is to the right of peak *Y* and that has the correct height.



See example of a correct response (dashed peak) above.

Chapter 09 Part 2: Rate Law

Part E

Part i

1 point is earned for the correct order and a valid justification.

The reaction is first order with respect to S8. Comparing trials 1 and 2, tripling [S8] while keeping [OH⁻] constant results in the tripling of the initial reaction rate.

OR

$$\frac{\text{rate}_2}{\text{rate}_1} = \frac{(k[\text{S}_8]^x [\text{OH}^-]^y)_2}{(k[\text{S}_8]^x [\text{OH}^-]^y)_1}$$

$$\frac{2.10\text{M/s}}{0.699\text{M/s}} = \frac{k(0.300)^x (0.0100)^y}{k(0.100)^x (0.0100)^y} \Rightarrow 3.00 = 3.00^x \Rightarrow x = 1$$

Part ii

1 point is earned for the correct concentration of OH⁻ and a valid justification.

Comparing trials 2 and 3, [S8] is kept constant and the initial reaction rate doubles. Since the reaction is first order with respect to OH⁻, the concentration of OH⁻ in trial 3 must be $2 \times 0.0100\text{ M} = 0.0200\text{ M}$.



0	1	2
---	---	---

Student earns 2/2 point(s)

Part i

1 point is earned for the correct order and a valid justification.

The reaction is first order with respect to S8. Comparing trials 1 and 2, tripling [S8] while keeping [OH⁻] constant results in the tripling of the initial reaction rate.

OR

$$\frac{\text{rate}_2}{\text{rate}_1} = \frac{(k[\text{S}_8]^x [\text{OH}^-]^y)_2}{(k[\text{S}_8]^x [\text{OH}^-]^y)_1}$$

$$\frac{2.10\text{M/s}}{0.699\text{M/s}} = \frac{k(0.300)^x (0.0100)^y}{k(0.100)^x (0.0100)^y} \Rightarrow 3.00 = 3.00^x \Rightarrow x = 1$$

Part ii

1 point is earned for the correct concentration of OH⁻ and a valid justification.

Comparing trials 2 and 3, [S8] is kept constant and the initial reaction rate doubles. Since the reaction is first order with respect to OH⁻, the concentration of OH⁻ in trial 3 must be $2 \times 0.0100\text{ M} = 0.0200\text{ M}$.

Part F

Chapter 09 Part 2: Rate Law

1 point is earned for a valid explanation that addresses the effect on energy of collisions.

The temperature was lower on the second day so the average kinetic energy of the reactant particles was lower. Therefore, there were fewer collisions between particles with sufficient energy to react.

1 point is earned for a valid explanation that addresses the effect on frequency of collisions.

Since the temperature was lower, the kinetic energy was lower and the average speed of the particles was lower. At the lower speeds, the reactant particles collided less frequently.



0	1	2
---	---	---

Student earns 2/2 point(s)

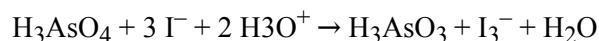
1 point is earned for a valid explanation that addresses the effect on energy of collisions.

The temperature was lower on the second day so the average kinetic energy of the reactant particles was lower. Therefore, there were fewer collisions between particles with sufficient energy to react.

1 point is earned for a valid explanation that addresses the effect on frequency of collisions.

Since the temperature was lower, the kinetic energy was lower and the average speed of the particles was lower. At the lower speeds, the reactant particles collided less frequently.

39.



The oxidation of iodide ions by arsenic acid in acidic aqueous solution occurs according to the stoichiometry shown above. The experimental rate law of the reaction is:

$$\text{Rate} = k[\text{H}_3\text{AsO}_4] [\text{I}^-] [\text{H}_3\text{O}^+]$$

According to the rate law for the reaction, an increase in the concentration of hydronium ion has what effect on this reaction?

(A) The rate of reaction increases.



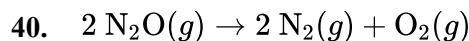
(B) The rate of reaction decreases.

(C) The value of the equilibrium constant increases.

(D) The value of the equilibrium constant decreases.

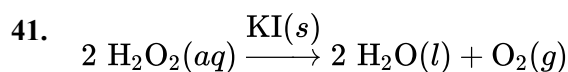
(E) Neither the rate nor the value of the equilibrium constant is changed.

Chapter 09 Part 2: Rate Law



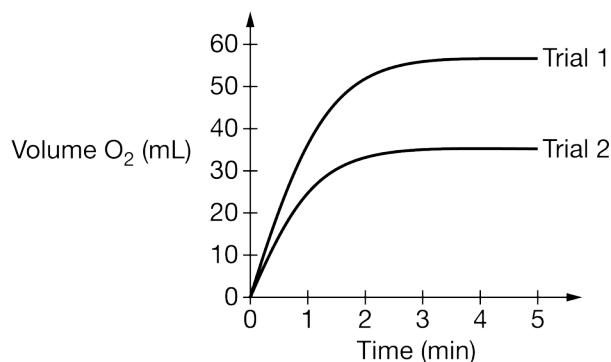
The rate law for the reaction represented above is experimentally determined to be $\text{rate} = k[\text{N}_2\text{O}]$. Which of the following best explains why increasing the initial concentration of $\text{N}_2\text{O}(g)$ at a constant temperature increases the reaction rate?

- (A) Molecular collisions become more frequent. ✓
- (B) There is a higher probability of effective molecular collisions.
- (C) The rate constant for the reaction increases.
- (D) The reaction mechanism changes to one that has a lower activation energy.



$$\text{Rate} = k[\text{H}_2\text{O}_2]$$

A student performed an experiment to study the factors that affect the rate of the first-order catalytic decomposition of $\text{H}_2\text{O}_2(aq)$, represented above.



The student monitored the volume of $\text{O}_2(g)$ produced over time as the reaction proceeded. The data from two trials are plotted in the graph above. Which of the following best explains the results of trial 2 compared with those of trial 1?

- (A) The reaction temperature in trial 2 was higher than it was in trial 1.
- (B) The volume of the reaction vessel was larger in trial 2 than it was in trial 1.
- (C) The student used a smaller amount of $\text{KI}(s)$ in trial 2 compared with that used in trial 1.
- (D) The concentration of $\text{H}_2\text{O}_2(aq)$ was lower in trial 2 than it was in trial 1. ✓